Project 10: RFID Reader

In this project we will use the Innovations ID-12 Radio Frequency Identification, or RFID reader to read unique identification numbers from cards, key fobs, stickers, and even little capsules. These tags have circuitry in them that when in proximity to a reader, powers up and broadcasts their unique 12-digit hexadecimal ID number. The reader transmits this data through its serial TX pin in 16-byte packets, one byte at a time. Because these tags are each unique, we can use them to inventory the beer in our fridge or to let only our cats in through our cat door.

The Innovations ID-12 is a common RFID reader that uses 125kHz tags and sends its data over a serial connection at a 9600 bps data rate. Other readers in the company's line include the ID-2 and ID-20, with similar functionality but different ranges and antenna requirements. The pins on the bottom of the device are an unusual spacing that is incompatible with our breadboard. So to make things easier for us, we are using a breakout board supplied by SparkFun Electronics that we have soldered some male pins to before soldering in the reader.

Hooking It Up

By using the breakout board from SparkFun, we can plug the reader into our breadboard easily enough, although, with only one row of open pins available, there is not much room to connect wires. So to make the connections, we are using a little area of the breadboard off to the side to connect common wires like +5VDC and Ground. Past the RFID reader, the only other part to connect is a piezo speaker that we used in our Noisy Cricket project to give us a little audible chirp whenever a card has been read, although an LED would work just as well. The wiring for this is stretching things a little, so double-check the schematic and drawings in Figures 10-3 and 10-4 to make sure the correct connections are being made.

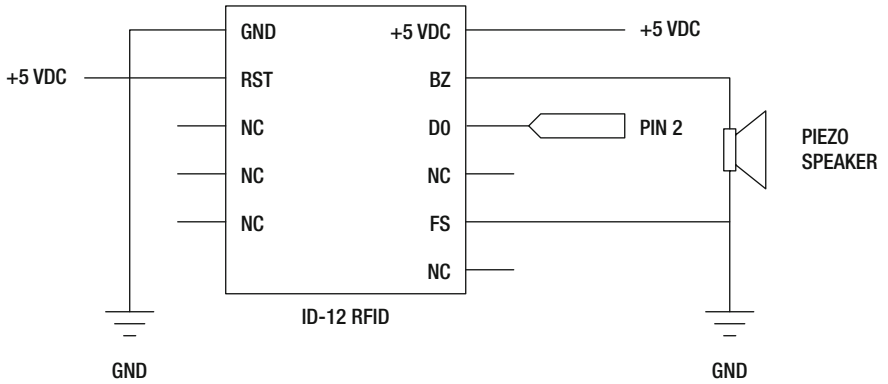

Figure 10-3. *RFID Reader schematic*